

HAKAI M1 — Wireless Gaming Mouse

Engineering Prototype Datasheet

Board Rev 5.3 (FINAL prototype) · Document Rev 1.1 · 2026-07-09 · HAKAI Engineering

General Description

The HAKAI M1 is a rechargeable 2.4 GHz wireless gaming mouse built around the Nordic nRF52833 SoC (integrated BLE / Nordic-proprietary 2.4 GHz radio) and the PixArt PAW3311DB-T2MU low-power optical gaming sensor. The entire system operates from a single 1.9 V rail, dictated by the sensor's 2.1 V absolute-maximum supply and enabled by the nRF52833's 1.7–3.6 V operating range — no level shifters anywhere in the design. Power comes from a single Li-ion cell with USB-C charging through a TI BQ24074 power-path charger.

All sensor wiring is verified against the PixArt datasheet (v1.0, 1A009EN); as of Rev 5.3 the sensor's **board mechanicals are also datasheet-exact** (Table 9, L0AJ-LSG1 lens column). All MCU support circuitry is verified against the Nordic nRF52833 Product Specification v1.7 and the PCA10100 reference design. The PCB is a shaped mouse plate with an open scroll-wheel bay and a split wheel assembly: rotation is sensed by an 11 mm encoder on the left bay wall, and the middle click is a fifth Kailh GM 8.0 microswitch on the right wall.

Production status: PLACED, NET-VERIFIED, UNROUTED. The included gerbers are for DFM / mechanical preview only — copper routing (in particular the RF path, which must copy Nordic's reference geometry) remains before production.

Key Features

- nRF52833-QIAA: 64 MHz Cortex-M4F, 512 KB flash, BLE 5 / 2.4 GHz proprietary radio, on-chip QDEC for scroll decoding
- PAW3311DB: up to 12,000 cpi, 300 ips, 35 g, 2.3 mA run / 3 μ A power-down (verified datasheet wiring: 3-wire SDIO + MISO/MOSI bridge; Table 9-exact cutouts, guide-post holes, pin holes)
- Single 1.9 V system rail (Torex XC6219B192MR, 240 mA), ferrite-isolated analog/RF sub-rail
- USB-C charge-only port at tail edge; BQ24074 power path: 445 mA charge, 497 mA input limit; BAT bypassed by 10 μ F ceramic at the pins (Rev 5.2)
- 5 $\times$ Kailh GM 8.0 (L, R, 2 $\times$ side, middle click) + 11 mm scroll encoder, all active-low
- High-side FET-gated battery divider on ADC AIN0 (Rev 5.2): true zero standby drain, ADC node clamped to GND when idle
- TI SWRA117D PCB inverted-F antenna in a copper-keepout zone
- SWD debug header (1.9 V logic) and 6 test points

Key Parameters

Parameter	Value	Note
System rail	1.9 V (1V9 / 1V9_A)	PAW3311 VDD abs max = 2.1 V governs
Battery	1S Li-ion 3.0–4.2 V, protected	JST-PH pin 1 = BAT+ (verify pigtail!)
Charge current / input limit	445 mA / 497 mA	ISET 2.0 k Ω , ILIM 3.24 k Ω , EN2=VSYS EN1=GND
Sensor SPI	3-wire SDIO, mode 3, $\leq$ 8 MHz	R23 bridge: 3.3 k @ 2 MHz, 1 k @ 4 MHz, 240 Ω @ 8 MHz

Scroll interface	quadrature A/B → nRF QDEC (P0.08/P0.11)	10 k pull-ups, 10 nF filters
Radio	2.4 GHz, single-ended ANT, Nordic ref match	1.0 pF / 4.7 nH / 1.2 pF / 2.2 nH + DNP tune cap
Board	70 × 110 mm envelope, 0.8 mm, 2-layer	thickness fixed by sensor lens stack
Battery sampling	~103 µA, only while P0.03 high	R25 47 k + 300 k divider (Rev 5.3)
Estimated cost	\$18.39 parts / \$24.89 incl. battery	qty-1..10 prototype estimates

MCU Pin Assignment (nRF52833-QIAA)

Function	Pin	Net	Function	Pin	Net
Sensor SCLK	P0.20	SPI_SCK	Left click	P0.04	BTN_LEFT
Sensor MOSI (via R23)	P0.22	SPI_MOSI	Right click	P0.05	BTN_RIGHT
Sensor MISO (SDIO direct)	P0.24	SENS_SDIO	Side back	P0.06	BTN_SIDE_BACK
Sensor NCS	P0.17	SENS_NCS	Side forward	P0.07	BTN_SIDE_FWD
Sensor MOTION (act-low)	P0.19	SENS_MOTION	Encoder A / B	P0.08 / P0.11	ENC_A / ENC_B
Battery ADC	P0.02 (AIN0)	VBAT_SENSE	Middle click	P0.12	ENC_SW
Divider enable (high-side FET pair)	P0.03	VBAT_EN	Status LED (via Q2)	P0.13	LED_STAT
Charger STAT / PGOOD	P0.14 / P0.15	STAT / PGOOD	Reset (UICR PSELRESET)	P0.18	nRESET
LFXO 32.768 kHz	P0.00/P0.01	XL1/XL2	HFXO 32 MHz	XC1/XC2	Y1 ± 12 pF

USB pins (VBUS, D±, DECUSB) are unconnected per Nordic's no-USB reference configuration — the USB-C port is charge-only. P0.09/P0.10 (NFC) unused. DEC values per Nordic: DEC1 100 nF, DEC3 100 pF, DEC4+DEC6 tied 1 µF + 47 nF, DC/DC 10 µH + 15 nH.

Connectors & Test Points

Ref	Type	Pinout
J1	USB-C 16-pin (GCT USB4105), tail edge	VBUS, GND, CC1/CC2 (5.1 k pulldowns), D± unconnected
J2	JST PH 2-pin battery	1 = BAT+ (design convention — no JST standard), 2 = GND
J3	SWD header 1×05, 2.54 mm	1V9, SWDIO, SWDCLK, nRESET, GND — probe VTref = 1.9 V
TP1–TP6	Test points (right ear)	SPI_SCK, SENS_SDIO, SPI_MOSI, SENS_MOTION, VBAT, VSYS

Scroll Wheel Assembly (split, type B)

The wheel, axle, and cradle are shell parts. The wheel's D-cut axle inserts into EC1 (11 mm rotation-only encoder, left bay wall — shaft points into the 24 × 38 mm open bay). Rotation produces 90°-phase-shifted A/B contact closures to GND, decoded by the nRF QDEC peripheral. Pressing the wheel tilts the cradle onto SW5 (Kailh GM 8.0, right bay wall, plunger end toward the bay) — the same switch family as the main clicks, giving M3 identical tactility. One switch

part number covers all five clicks. Verify EC1 pin/bracket spacing against the chosen vendor drawing (TTC/F-switch) before fab.

Mechanical

- Shaped mouse plate: switch "ears" flanking the open wheel bay, chamfered corners, narrowed tail; 3× M3 mounting holes
- PCB thickness 0.8 mm (PAW3311 lens stack requirement); sensor optical center at (135, 106) board coordinates
- Sensor board features **exact per PixArt Table 9** (L0AJ-LSG1 SMT lens set; lens-to-surface Z = 2.20–2.60 mm): full aperture cutout with optical-axis notch, R0.90 × 2.00 stadium LED cutout, 2× lens guide-post holes (2.0 × 1.6, R0.40 — lens self-aligns and can be heat-staked), Ø0.80 sensor pin holes, IR-LED pads at the Table 9 positions (custom footprint; confirm vs final LED P/N)
- Kailh switch pattern verified against official Omron D2F/D2FC drawings: 3× Ø1.2 terminal holes at 5.08 pitch; switch bottoms are flat — **no locating-boss holes exist in the D2F family** (the side-face holes in the drawings sit 1.5 mm above the PCB and are chassis-pin features)
- Antenna zone (tail, right) kept copper-free on both layers

Verification Summary (all machine-checked)

Check	Result
Schematic ERC	0 violations
Netlist vs design intent	45 signal nets + 6 rails exact-match
PCB pad-by-pad electrical audit	306 netlist assignments vs 319 pads: 0 mismatches
PCB DRC (with filled GND pours)	0 error-severity violations (excl. intentional unrouted ratsnest)
Pad containment in shaped outline	all pads inside
BOM ↔ netlist reconciliation	95 refs == 95 components
Cost arithmetic	line items sum exactly to stated totals
Drill files vs board	PTH 44 + NPTH 5 holes, exact match

Reports are shipped in the `verification/` folder. Whitelisted board-side extras (all verified): nRF die pad → GND, USB shield → GND, 32 MHz crystal shell pads → GND, antenna ground stub → GND, encoder bracket tabs (mechanical).

Gerber Package

`GERBERS_hakai_final_PROTO.zip` — F/B copper (GND pours filled), F/B mask, F/B paste, F/B silkscreen, board outline, gerber job file (X2 revision = 5.3), and separated PTH/NPTH Excellon drill files.

These gerbers describe the UNROUTED prototype: placement, pours, cutouts, and holes are final-candidate; signal routing is absent. Suitable for fab-house DFM review, mechanical fit checks, and shell prototyping — not for electrical production.

Before Production (open items)

1. **Route the board.** RF path (ANT → matching → antenna): copy Nordic PCA10100 trace/ground geometry verbatim, 50 Ω, stitching vias; VNA-tune the DNP antenna cap (C24). Keep SPI/battery routing away from RF.
2. Verify EC1 encoder pin/bracket spacing against the exact vendor drawing.
3. Match shell cradle press travel to the Kailh SW5 plunger height; GM 4.0 is a drop-in lighter option.
4. Confirm D2's pads against the reflow pad recommendation of the actual IR emitter purchased (PixArt names the P/N for the L0AJ-LSG1 set).
5. Confirm battery pigtail polarity with a DMM; use a protected cell.
6. After routing: re-run DRC, then emissions pre-scan (FCC/CE) with the tuned antenna.

Firmware Bring-Up Notes

SPI mode 3 at ≤ 2 MHz initially (matches fitted R23 = 3.3 k). Sensor wake-from-shutdown requires RESET, then 50 ms (tMOT-RST) before valid motion. Scroll via QDEC on P0.08/P0.11. nRESET is P0.18 via UICR PSELRESET (both registers). DC/DC REG1 enabled (inductors fitted). Battery reading: drive P0.03 high, sample AIN0 (VBAT/3 ≤ 1.4 V), release — ~103 μA only while sampling; true zero standby drain (AIN0 held at GND by R9 when idle).

Revision History

Rev	Change
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1.1	Original prototype schematic (rectangular concept, 12-pin sensor symbol) — audited, 13 findings
2.x	Corrected design; sensor wiring re-verified against PixArt datasheet → single 1.9 V rail, 3-wire SDIO bridge
3.0	Schematic layout cleanup (no overlaps)
4.x	Shaped mouse-plate PCB: open wheel bay, cutouts, mounting holes; full pad-level electrical audit
5.0	Split wheel: EC1 encoder (left wall) + tact middle click (right wall)
5.1	Middle click upgraded to 5th Kailh GM 8.0; BOM consolidated
5.2	Electrical fixes from full datasheet audit: C36 BAT bypass (BQ24074 Table 7-1); battery divider re-topologized to high-side gating (Q3 AO3401A + R25) — AIN0 can no longer float above VDD, true zero standby drain
5.3	Sensor mechanicals datasheet-exact (Table 9: full aperture + notch, LED stadium, guide-post holes, Ø0.80 pins, custom IR-LED pads); Kailh pattern verified vs official Omron drawings (flat bottom, no boss holes — item closed); R25 → 47 k (2× gate-threshold margin); FINAL prototype package

HAKAI M1 Engineering Prototype Datasheet · Rev 1.1 · Generated from the verified KiCad project (hakai_mouse_final). Component cost estimates are qty 1–10 prototype figures (2026 USD) — verify current distributor pricing before ordering. This document describes an unrouted engineering prototype and is not a production specification. PAW3311DB electricals and Table 9 mechanicals summarized from PixArt datasheet 1A009EN v1.0; nRF52833 values from Nordic PS v1.7 / PCA10100; switch pattern from official Omron D2F/D2FC datasheets and the Kailh GM 8.0 drawing.